# Response to the Referees for SYNT-D-26-00214

1. clearify the real/apparent definition, 2. clarify the meaning of "vanishes" for the transverse Doppler effect, 3. claifiy the role of Bell's type dynamical explanations

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**Reviewer 1**

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**Reviewer 2**

**Reviewer (2A)** — First, it is claimed that length contraction and time dilation, in the eyes of Einstein, are "apparent" effects because they "vanish" for the co-moving observer, but are nonetheless "real" since they "cannot be eliminated in all frames at once" (pp. 2, 22, footnote 8). This latter criterion of reality is almost surely NOT what Einstein had in mind.

**Reply**: My explanation of the the distinction between *real* and *apparent* was was unclear. It is introduced as an exegetical reconstruction of Einstein's own usage. The clearest formulation occurs in his 1915 letter to Lorentz:

> Regarding the erroneous view that the Lorentz contraction was 'merely apparent,' [*scheinbar*] I am not free from guilt, without ever having myself lapsed into that error. It is real [*wirklich*], i.e., measurable with rods and clocks, and at the same time apparent [*scheinbar*] to the extent that it is not present for the co-moving observers (Einstein to Lorentz, Jan. 23, 1915; **CPAE**).

Accordingly, the paper does not claim that an effect is real because it *cannot be eliminated in all frames at once* (this applies merely to the case of Ehrenfest paradox, see 2B). Rather, it reconstructs Einstein's own characterization:

- apparent = absent for the co-moving observer.
- real = experimentally measurable,

The transverse Doppler effect illustrates precisely this point. A spectrometer co-moving with the emitting ion records its proper frequency, so that no shift is observed in that experimental arrangement. A spectrometer in relative motion, however, measures the shifted frequency. Thus the effect is relational, but this is exactly what Einstein means by calling it simultaneously *real* and *apparent*. The fact that the shift disappears in one observational arrangement does not make it illusory; it remains experimentally measurable in another (see also my response to 2C).

**Reviewer (2B)** — Applications of length contraction and time dilation can, in the appropriate circumstances, involve non-classical empirical predictions, such as the breaking of the string in Bell's two-rockets thought-experiment (footnote 1), stresses in a rotating disk (p. 2), the experimental confirmation of the transverse Doppler shift (section 2), and time retardation in the twin's "paradox" scenario (not mentioned in the paper). These predictions of special relativity are ones that ALL observers will agree on, including the observer in the relevant co-moving frame(s). It is in this sense that length contraction and time dilation are supposedly real. Indeed in footnote 1, the author explicitly states that the examples of "stresses and the possible structural failure of the rope (or of the rotating disk)" mean that length contraction is physically real for Einstein — these are coordinate-independent predictions.

**Reply**: We agree that the original formulation did not sufficiently distinguish the local and global senses in which a relativistic effect may be said to "disappear" for a co-moving observer. This distinction has now been made explicit.

In the local case, a particular length contraction or time dilation can be made to disappear by passing to the rest frame of the relevant material element or clock. In that operational sense the effect is "apparent": the co-moving observer measures the proper length, proper time, or proper frequency. It is nevertheless "real" in Einstein's sense, because it is measurable with rods and clocks in a non-co-moving arrangement.

In extended or non-inertial situations, however, the local rest frames need not assemble into a single global co-moving inertial frame. This is what happens in the rotating disk: each rim element has an instantaneous rest frame, but the collection of these frames cannot be turned into one global rest frame for the disk. The same point applies, mutatis mutandis, to the twin paradox: the travelling twin is locally at rest in successive instantaneous frames, but these cannot be combined into a single inertial frame covering the whole journey. In such cases the effect may disappear locally, but it cannot be eliminated globally, and the failure of globalization has observer-independent consequences, such as stresses in the disk or differential aging in the twin case.

The transverse Doppler effect occupies an important place in the paper because, unlike the early discussions of length contraction, it offered Einstein a concrete empirical route for testing time dilation. A co-moving spectrograph records the proper frequency of the emitting ion, so the shift disappears in that local observational arrangement. A non-co-moving spectrograph, however, records the displaced spectral line. This made time dilation experimentally accessible in a way that length contraction, at the time, was not.

**Reviewer (2C)** — The transverse Doppler effect concerns the claim is that the frequency of emission of a moving ion, the emission being in the transverse direction, is different from the proper frequency as defined relative to the co-moving frame (which is the same as the proper frequency defined relative to K). It is arguably misleading to say that the effect "vanishes" relative to the co-moving frame. The effect might be described as a relation between two inertial frames, as is brought out on p. 16. Better put, it is a relation between proper frequency relative to K and that related to an ion moving relative to K. In the author's scenario, relative to K' this relation does not exist— there is a single "clock" and one cannot even define the Doppler shift, and so it cannot have zero value (if that is what the author means by "vanishing"). However, in the reversed scenario in which the observer at rest in K' is looking at emitting ions at rest relative to K, then the transverse Doppler shift is defined and the effect is the same as in the original case.

**Reply**:

I did not realize that my use of the term 'vanish' was indeed ambiguous. By saying that the effect 'vanishes' for the co-moving observer, I use the term in an operational sense. The relevant question is what is recorded by measuring devices at rest with the system under investigation. A rod measured by co-moving rods has its proper length; a clock compared with co-moving clocks has its proper rate; an emitting ion observed by a co-moving spectrograph displays its proper frequency. In such an arrangement, the observable difference that defines the effect in a non-co-moving frame is not registered.

Thus, in the transverse Doppler case, the claim is not that the Doppler shift is an intrinsic property of the ion which somehow takes the value zero in its rest frame. Rather, a spectrograph co-moving with the emitting ion records the normal rest line, $\nu = \nu_0$, whereas a spectrograph in transverse relative motion records the shifted line, $\nu = \gamma^{-1}\nu_0$. This is the precise sense in which the shift “vanishes” for the co-moving observer. It is absent in the co-moving observational arrangement, but experimentally measurable in the non-co-moving one.

**Reviewer (2D)** — The second point I want to make is more important. The author attempts to show, through an analysis of an important historical episode related to the testing of time dilation, that there is a confusion in the claim by Brown and others that such a "kinematical" effect (to use Einstein’s original wording) has a dynamical underpinning, despite Einstein’s belated recognition, endorsed by the author, that rods and clocks are physical systems governed by dynamical laws. Essentially, the author’s argument is this. A dynamical account of, specifically, the transverse Doppler effect in special relativity, under plausible assumptions, would explain why all atomic clocks are identical when compared in the same rest frame, but no more. Explaining the transverse frequency shift requires, according to the author, more than that: it further requires time dilation regarded as a kinematical, not dynamical effect (pp. 3, 24).

Here are some objections to this view.

Bell showed in his famous 1976 paper on teaching special relativity that a purely dynamical (pre-quantum) model of a moving single-electron atom has time dilation unequivocally emerging from dynamical assumptions (largely Maxwell electrodynamics) in this specific model. Recall his point: the special merit of this approach "is to drive home the lesson that the laws of physics in any one reference frame account for all physical phenomena, including the observations of moving observers" (Bell 1976). The author’s appeal to irreducible kinematics in fully accounting for the transverse Doppler effect seems to be in direct conflict with this important insight.

Furthermore, Bell makes a nice analogy with the parallel roles of thermodynamics (which gave Einstein the methodological template for his 1905 relativity paper) and statistical mechanics. He wrote:

"If you are, for example, quite convinced of the second law of thermodynamics, of the increase of entropy, there are many things that you can get directly from the second law which are very difficult to get directly from a detailed study of the kinetic theory of gases, but you have no excuse for not looking at the kinetic theory of gases to see how the increase of entropy actually comes about. In the same way, although Einstein’s theory of special relativity would lead you to expect the FitzGerald contraction, you are not excused from seeing how the detailed dynamics of the system also leads to the FitzGerald contraction." (Bell 1976, as referenced in the paper.)

Of course the same can be said of time dilation. The key point is this. Naturally, Einstein used time dilation as established in the "kinematical" part of his 1905 paper in his treatment of the transverse Doppler effect. This was because the full dynamical theory of moving Ions and their emission properties was not available to him, and even today would be very complicated. The beauty of the kinematical approach was that it did not rely on the fine-grained details of the dynamics, as the author acknowledges. It is analogous to the way thermodynamics leads to an entropy non-decrease law without specifying all the complicated details that would be required in a purely mechanical analysis of the relevant system. But as the current author concedes, special relativity is a set of constraints on theories. Indeed, it is much the way that thermodynamics is. Because something is hard to prove in what Einstein called a constructive (dynamical) theory of matter and radiation — even if it available —, while at the same amenable to a simple principle-theory treatment, does not mean that the latter is fundamental, or in principle irreplaceable. The author takes the lesson of the transverse Doppler effect to be that "kinematic and dynamics, from a strictly logical point of view, cannot be confronted with experience separately but only as a whole. From the point of view of explanation, however, the division of labour within this whole remains intact." (pp. 23, 24). The analogous, erroneous, claim would be that certain phenomena in the behaviour of gases, for example, can only be explained in principle by an irreducible combination of thermodynamics and statistical mechanics.

**Reply**: We thank the referee for raising this important point. We agree that Bell's 1976 discussion must be taken seriously, and we have revised the manuscript to clarify the target of our argument. The paper does not deny that one can construct a detailed model calculation in which a material system, described from a single inertial frame, exhibits the Lorentz-contracted or time-dilated behavior expected by special relativity. Bell's example is valuable precisely because it shows how relativistic behavior can be realized by concrete physical systems governed by appropriate dynamical laws.

The point at issue, however, is how such a calculation is to be interpreted.

- a If Bell's procedure is read as a genuinely constructive explanation of time dilation in the strong dynamical sense, then the motion of the atom relative to some privileged state of rest must make the moving atom dynamically different from an atom at rest. This is close to the neo-Lorentzian strategy pursued by Herbert E. Ives in connection with the Ives-Stilwell experiment on the rate of a moving atomic clock. On that interpretation, the transverse Doppler effect is explained by a real dynamical alteration of the moving atomic system. But this is not the constructive program usually intended by defenders of a dynamical approach within special relativity, since it reintroduces precisely the preferred-frame structure that Einstein's theory was designed to avoid.
- If, by contrast, Bell's model is interpreted within Einsteinian special relativity, then the relevant dynamical laws are Lorentz covariant. A moving atom is not dynamically different from an atom at rest; it is the Lorentz-transformed counterpart of the same kind of physical solution. In that case the dynamics explains why stable atomic clocks exist, why atoms of the same kind have the same proper frequency when compared at rest, and why such systems can serve as clocks. But the transverse Doppler factor itself follows from the Lorentz-transformational relation between the proper time of the emitting ion and the time coordinate of the non-co-moving observational frame. The dynamical model realizes the relativistic kinematics; it does not replace it with an independent dynamical alteration of the clock.

We have therefore revised the manuscript to avoid suggesting that a Bell-style dynamical calculation is impossible or unimportant. The intended claim is narrower: within Einsteinian special relativity, a dynamical account of the atom explains the identity and stability of the atomic clock, while the transverse Doppler shift is explained by the Lorentz transformation relating the clock's proper time to the time coordinate of a non-co-moving frame. A stronger dynamical explanation of the shift itself is available only by moving toward a Lorentzian or Ives-style interpretation, according to which moving atoms undergo a real dynamical modification relative to a preferred frame. I do not see the third alternative of a dynamical explanation within relativity, without retuirn to the ether theory.

**Reviewer (2E)** — It stated that the Lorentz transformations make their "perfect symmetry in $x$ and $t$ explicit". Is it perfect?

**Reply**: I agree that the formulation was too strong. I have replaced it with a more cautious statement: the Lorentz transformations make explicit the reciprocal mixing of spatial and temporal coordinates, especially when the time coordinate is written as $ct$.

**Reviewer (2F)** — It is claimed that Einstein was the first to see time dilation as a genuinely new physical effect. But Larmor and Lorentz independently had a clear premonition of time dilation in 1897 and 1899 respectively, though it is not clear that they thought of it as a universal property of ideal clocks.

**Reply**: I thank the referee for this correction. I have qualified the claim. The revised text now acknowledges that Larmor and Lorentz had obtained related formal results before Einstein. The point

I intended to make is narrower: Einstein's distinctive contribution was to interpret clock retardation as a universal consequence of the new kinematics, applying to any suitable clock, rather than as a feature of a particular electromagnetic model.

**Reviewer (2G)** — The coordinate time interval in $K$ between two successive ticks of a moving clock cannot be measured by a single clock at rest in $K$, since the ticks occur at different spatial points in $K$. This risks inconsistency with the claim that the moving clock runs "more slowly" than an identical clock at rest in $K$.

**Reply**: I agree. I have revised the relevant passage to make explicit that $\Delta t$ is read off by the synchronized lattice of clocks at rest in $K$, whereas $\Delta t'$ is the proper time measured by the single clock at rest in $K'$. I have also replaced the potentially misleading phrase that the moving clock "runs more slowly" with a formulation in terms of the longer coordinate-time interval assigned in $K$ to successive ticks of the moving clock.

**Reviewer (2H)** — The manuscript seems to treat the invariance of the number $N$ of oscillations as a substantive assumption based on the relativity principle. But $N$ is a count and is independent of all coordinate frames by its nature.

**Reply**: I agree that the previous wording was misleading. I have revised the manuscript throughout to distinguish the trivial invariance of the count $N$ from the substantive physical premise. The point is not that $N$ itself is made invariant by the relativity principle, but that uniform motion is assumed not to alter the intrinsic oscillatory process of the emitter. Thus identical ions, when compared at rest, possess the same proper frequency $\nu_0 = N/\Delta t'$. The transverse Doppler effect then arises because the same count $N$ is referred to different time intervals, $\Delta t'$ and $\Delta t = \gamma \Delta t'$, in the two inertial descriptions.

**Reviewer (2I)** — Ritz is described as criticizing the abandoning of absolute time and rigid bodies in the Lorentz–Einstein theory. But this theory does not abandon idealized rigid bodies.

**Reply**: I agree. I have revised the wording to clarify that Ritz was criticizing the abandonment of absolute time and of the classical notion of absolutely rigid bodies, not the use of idealized rods or rigid bodies as measuring instruments within the relativistic framework.

**Reviewer (2J)** — Typos: first line p. 16; third line final paragraph p. 18; equation in last line of first paragraph of section 4, p. 20.

**Reply**: I thank the referee for pointing these out. I have corrected these typographical errors in the revised manuscript.

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**Reviewer 3**

The present submission contributes to the literature on the relation between dynamical and kinematical considerations in special relativity. Specifically, it disputes a narrative to the effect that Einstein initially treated time dilation as a merely apparent coordinate artifact but sought a dynamical explanation on which time dilation reflects real changes, thereby suggesting a precursor to today's dynamical view. To challenge this narrative, the paper reconstructs in detail the role of the kinematical/dynamical distinction in Einstein's treatment of the transverse Doppler effect. It argues that Einstein was interested not in a dynamical explanation of the transverse Doppler effect but rather in a dynamical backing for assumptions going into experimental testing of the effect.

This paper is timely because it concerns debates over kinematical and dynamical explanation in special relativity. The topic is also relevant to those with historical interests in the special theory of relativity. The historical analysis exhibits detailed engagement with the primary sources and substantial work in synthesizing them. It successfully makes its conceptual case on the basis of this analysis.

Minor typos: 2: "as [shown] by the stresses in Ehrenfest's disk" 9: "[in] November of 1905" 23: "which are in turn required to invariant with [respect to] the Lorentz transformations"